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Numerical modeling of nitrogen transport and loss during laser powder bed fusion of high nitrogen steels

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ABSTRACT

High Nitrogen Steels represent a highly promising material class for Laser Powder Bed Fusion, but their processing is challenged by significant nitrogen loss due to a drop in solubility upon melting. To systematically investigate the underlying degassing kinetics and nitrogen transport mechanisms, this study establishes a high-fidelity computational fluid dynamics model utilizing the Volume of Fluid method. The continuous species conservation framework accounts for convection-dominant transport, integrating a temperature-dependent nitrogen solubility limit and an interfacial mass transfer sink term at the free melt pool surface. The numerical model is successfully calibrated and validated against single-track experiments on Fe-15Cr-0.3C-N steel substrates across varying surface energy densities, demonstrating good agreement with both experimental melt pool geometries and a spatial nitrogen concentration profile measured via Wavelength Dispersive X-ray Spectrometry. The results reveal that the final nitrogen distribution across the melt pool depth is fundamentally governed by intense keyhole degassing coupled with complex convective transport. Marangoni convection drives nitrogen depleted melt backwards, inducing steep surface concentration gradients, while recirculation vortices transport the degassed melt deeper into the substrate. Using a novel Nitrogen Retention Index, parametric analysis demonstrates that higher surface energy densities prolong active surface degassing, thereby intensifying nitrogen loss.

1. Introduction

The Laser Powder Bed Fusion (PBF-LB/M) process iteratively spreads thin layers of metal particles, which are melted with a laser heat source and solidified layer by layer offering design freedom and rapid prototyping capabilities. In recent years, the PBF-LB/M process gained more and more attention due to the high accuracy (±0.1 mm), wide variety of materials and unique mechanical properties making the process applicable in fields such as aerospace [1], medical and dental technology [2], energy and process engineering [3] and chemical industry [4]. Against this background, N-alloyed stainless steels represent a particularly promising material class for additive manufacturing by PBF-LB/M, as nitrogen provides a unique pathway to tailor both mechanical performance and corrosion resistance [5,6]. Similar to carbon, nitrogen occupies the interstitial lattice sites in iron, thereby causing a lattice distortion that results in solid-solution strengthening, providing high strength while avoiding the alloying with carbon [7]. In contrast to carbon, nitrogen promotes short-range ordering of substitutional elements within the austenite, enhancing chemical homogeneity at hardening temperatures and reducing the local driving force for precipitation. Furthermore, nitrogen slows the kinetics of chromium-rich carbide formation, thereby mitigating the development of chromium-depleted regions. Consequently, a larger fraction of chromium remains available for the formation of a Cr-rich M_2O_3 passive layer, resulting in improved resistance to localized corrosion phenomena, including intergranular, pitting and crevice corrosion [7–9].

These advantages justify the use of high nitrogen martensitic stainless steels, which have been developed over recent decades and whose conventional processing by Pressurized Electroslag Remelting is well understood. The solubility of nitrogen in Fe–Cr–C system is limited and strongly depends on the phase present. In the solid austenitic phase the maximum solubility is larger than in the ferritic phase due to its larger octahedral lattice sites [10]. As soon as the steel melts, the maximum solubility significantly drops leading to the formation of gas pores and nitrogen loss at the free surfaces of the molten steel. Here,

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PBF-LB/M with high cooling rates in the order of 10^5–10^6 K/s can reduce the nitrogen loss due to rapid solidification. Boes et al. [11] showed that using nitrogen as a process gas during gas atomization effectively introduces up to 0.18 wt-% of nitrogen into X30CrMoN15-1 martensitic stainless steel powder. While they observed a slight reduction in nitrogen content to 0.14 wt-% during the PBF-LB/M process, the resulting parts maintained a high density and a predominantly martensitic microstructure with significant amounts of retained austenite. More recently, Becker et al. [12] published a novel process route for the additive manufacturing of high nitrogen martensitic stainless steel by combining gas-atomization with subsequent nitrogen gas nitriding. They overcame the limitations of the maximum solubility of the molten particles during gas atomization [11], resulting in a target nitrogen content of 0.71 wt-% by mixing gas-atomized powder with nitrided powder. The rapid solidification rates of the PBF-LB/M process successfully retained 0.474 ± 0.006 wt-% nitrogen in the as-built state, which is still above the thermodynamic equilibrium solubility of nitrogen. Next, Hot Isostatic Pressing (HIP) as post-processing closed PBF-LB/M induced porosity while internal rapid quenching was applied for martensitic hardening resulting in a high nitrogen martensitic stainless steel with a final nitrogen content around 0.245 ± 0.004 wt-%. The nitrogen loss during HIP was attributed to the low nitrogen partial pressure within the atmosphere, leading to denitriding of the samples. Despite these advancements, the underlying mechanisms of nitrogen transport and the precise kinetics of nitrogen loss during the PBF-LB/M and HIP are not yet fully understood and remain exceptionally difficult to characterize experimentally in-situ.

Given these experimental limitations, numerical modeling provides a powerful approach to systematically investigate the underlying mechanisms of nitrogen transport and the precise kinetics of nitrogen loss. Pitkälä et al. [13] developed an empirical-thermodynamic-kinetic model for industrial-scale converters, demonstrating that nitrogen loss follows a second-order reaction mechanism due to the required recombination of nitrogen atoms to molecular N_2 at the interface. While these models successfully predict nitrogen content based on gas flow rates and melt composition, they are primarily adapted for large-scale metallurgical processes like the argon oxygen decarburization converter. In alignment with large-scale metallurgical processes, numerical studies have been extended to industrial vacuum tank degasers to understand the complex kinetics of gas removal. Yu et al. [14] developed an integrated Computational Fluid Dynamic (CFD) model using the Euler-Euler approach to simulate the removal of nitrogen and hydrogen from medium carbon steel. The model specifically addresses the degassing, which is controlled by the mass transfer of dissolved nitrogen in the liquid steel and the chemical reaction kinetics at the gas-steel interface. To accurately capture the nitrogen transport at the free surface and bubble interfaces, the researchers implemented an activity coefficient model based on the liquid phase substitutional solution. The mass transfer coefficient in the liquid phase is determined via an eddy-cell expression [15], which accounts for surface renewal driven by turbulent kinetic energy dissipation in the gas-stirred melt. In alignment with these studies, Dong et al. [16] investigated nitrogen kinetics during the vacuum circulation process, using an optimized transport model based on a combined thermodynamic and kinetic approach. The mathematical framework partitions the reactor into three simultaneous zones governed by pressure-dependent equilibrium. Kinetically, the model defines the interfacial chemical reaction, specifically the adsorption and dissociation of N_2 molecules, as the rate-determining step. The degassing rate is calculated using a second-order rate equation that incorporates the activity of vacant surface sites. Numerical investigations for the PBF-LB/M process have been carried out for vaporization-induced composition change [17,18], in-situ alloying of multi-materials [19,20], thermo-viscoelastic response and residual stresses of printed parts [21], and the vapor dynamics within the gas phase [22,23]. To capture multi-component fluid dynamics, Chouhan et al. [19] developed a transient computational framework coupling a Volume of Fluid (VoF) method with a species conservation equation. Their study on the in-situ alloying of dissimilar metals revealed that Marangoni convection and recoil pressure strongly govern the elemental mixing. To support their numerical findings, the simulations were validated against single-track experiments, demonstrating good agreement for macroscopic melt pool dimensions via optical microscopy and for microscopic chemical inhomogeneities using Wavelength Dispersive X-ray Spectrometry (WDS). They concluded that the rapid solidification inherent to the PBF-LB/M process prevents complete chemical homogenization, leaving a heterogeneous elemental dispersion. Expanding on process-induced concentration gradients, Wang et al. [18] recently introduced a thermal-fluid model specifically accounting for the evaporative loss of highly volatile elements. By implementing an evaporation-driven mass sink at the free surface, they demonstrated how keyhole fluctuations and convective flows transport depleted liquid. This mechanism results in a heterogeneous, 3D cirrus-shaped composition distribution within the melt pool that directly degrades local mechanical properties. To ensure physical accuracy, their model was validated using Electron Probe Micro Analysis to map spatial element distributions. While these high-fidelity models successfully capture liquid-phase mixing and surface vaporization, the specific thermodynamics of dissolved nitrogen has not yet been addressed. To the best of the authors knowledge, no simulation studies have been published on super-saturated nitrogen outgassing for high nitrogen martensitic stainless steel during the PBF-LB/M processing.

While existing high-fidelity PBF-LB/M models have successfully addressed general solute transport and evaporation-induced composition changes, the specific thermodynamics of dissolved nitrogen and highly super-saturated melt pools have not yet been investigated. To bridge this gap, this work develops a novel transient high-fidelity model to systematically investigate the mechanisms of nitrogen transport and the kinetics of super-saturated outgassing during PBF-LB/M for high nitrogen steels. Central to this framework is a continuous species transport equation that explicitly accounts for super-saturation-induced desorption at the free surface of the melt pool. This model represents the first PBF-LB/M framework to directly integrate thermodynamic calculations for both the temperature-dependent nitrogen saturation limit and the driving forces of gas desorption. To ensure physical accuracy, the numerical framework is validated against novel, dedicated single-track experiments performed on highly super-saturated Fe-15Cr-0.3C-N bulk substrates using WDS. This approach provides a fundamental analysis of local degassing rates and reveals how complex convective transport, specifically Marangoni convection and recirculation vortices, governs the final nitrogen distribution within the melt pool. Finally, a parametric study introduces a Nitrogen Retention Index (NRI) to quantify the impact of processing parameters, offering a novel predictive tool for the development of high-performance N-alloyed stainless steels.

2. Mathematical formulation

A transient micron-scale high-fidelity model is employed to conduct single-track simulations within stainless steel substrates characterized by nitrogen contents exceeding the solubility limit. The model utilizes the VoF method, incorporating coupled heat transport and transfer mechanisms, while the laser energy input is simulated via a ray-tracing algorithm. Subsequently, nitrogen transport is resolved using a continuous species equation that accounts for convection, diffusion, and mass loss at the free surface of the melt pool.

2.1. Momentum and mass transport

The initial numerical configuration comprises a metallic phase with a uniform nitrogen concentration (N) and a gaseous nitrogen phase (N_2). Upon the onset of melting induced by the laser energy input, the VoF approach is utilized to track the metal-gas interface. Within this framework, the scalar field variable γ is employed to distinguish

between the metallic phase ($\gamma = 1$) and the gaseous phase ($\gamma = 0$), resulting in the following transport equation [24]:

$$\frac{\partial \gamma}{\partial t} + \nabla \cdot (\boldsymbol{u}\gamma) + \nabla \cdot \left((1-\gamma)\gamma \boldsymbol{u}_r\right) = 0\,. \tag{1}$$

In this formulation, $\boldsymbol{u}$ denotes the velocity vector representing the advection of the volume fraction driven by fluid motion, while $\boldsymbol{u}_r$ signifies the artificial compression velocity employed to sharpen the interface within the VoF framework. This term is essential for mitigating numerical diffusion.

The fluid flow within both the liquid melt pool and the gaseous phase is assumed to be incompressible. Consequently, the mass conservation equation reduces to the following simplified form:

$$\nabla \cdot (\boldsymbol{u}) = 0\,. \tag{2}$$

It should be noted that metal evaporation and nitrogen degassing are neglected in the mass conservation equation, as their contribution to the overall mass balance is considered negligible. The modified momentum equation for laminar, incompressible flow within both the melt pool and the gaseous phase is formulated as follows [19]:

$$\begin{aligned} &\frac{\partial (\rho \boldsymbol{u})}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}\boldsymbol{u}) - \nabla \cdot (\mu \nabla \boldsymbol{u}) = \\ &-\nabla p_{rgh} - \boldsymbol{g} \cdot \boldsymbol{h} \nabla \rho + \rho \beta \boldsymbol{g} \left(T - T_{ref}\right) + \boldsymbol{F}_D + \boldsymbol{F}_I\,. \end{aligned} \tag{3}$$

In this context, ρ, μ, p_{rgh}, $\boldsymbol{g}$, $\boldsymbol{h}$, β and T_{ref} denote the density, dynamic viscosity, modified pressure, gravitational acceleration, fluid column height vector, thermal expansion coefficient, and reference temperature, respectively. To account for natural convection driven by thermal gradients while maintaining this incompressible framework, the Oberbeck-Boussinesq approximation is explicitly applied. Under this approximation, density variations are neglected in the continuity equation (see Eq. (2)) and inertial terms but are retained within the buoyancy source term $\rho\beta\left(T - T_{ref}\right)$ in the momentum equation. Although the extreme temperature range within the keyhole exceeds the conventional validity limits of the Boussinesq approximation, its application remains highly justified in this specific process regime. Within the PBF-LB/M melt pool, fluid flow is predominantly governed by the intense Marangoni convection and recoil pressure at the free surface. Because buoyancy effects are of secondary importance compared to these dominant interfacial forces, any density-related inaccuracies introduced by this approximation have a negligible impact on the overall predicted melt pool dynamics [25,26]. Furthermore, $\boldsymbol{F}_D$ represents the Darcy-like source term based on the Kozeny–Carman equation, which accounts for the flow resistance within the mushy zone. This term effectively reduces the momentum at the solid–liquid interface, ensuring that the fluid velocity within the solid phase remains zero [27]. Interfacial forces at the gas-liquid boundary, including surface tension, Marangoni convection, and recoil pressure, are incorporated via the source term $\boldsymbol{F}_I$ [26]:

$$\boldsymbol{F}_I = \left(\sigma \kappa \hat{n} + \frac{d\sigma}{dT}\left[\nabla T - \hat{n}\left(\hat{n} \cdot \nabla T\right)\right] + \hat{n}\left(P_v\right)\right) |\nabla \gamma| \frac{2\rho}{\rho_{metal} + \rho_{gas}}\,. \tag{4}$$

In this formulation, σ, κ, $\hat{n}$, $\frac{d\sigma}{dT}$ and P_v represent the surface tension coefficient, the interface curvature, the unit normal vector to the interface, the temperature coefficient of surface tension, and the recoil pressure, respectively. The latter is derived from the Clausius-Clapeyron relation as follows:

$$P_v = 0.54 P_0 \exp\left[\frac{L_v M \left(T - T_v\right)}{R T T_v}\right], \tag{5}$$

where P_0, L_v, M, T_v and R denote the ambient pressure, the latent heat of vaporization, the molar mass, the boiling temperature of the metal, and the universal gas constant, respectively.

2.2. Heat transport and ray tracing

Energy conservation is governed by the following modified heat transport equation [19]:

$$\frac{\partial \left(\rho C_p T\right)}{\partial t} + \nabla \cdot \left(\rho \boldsymbol{u} C_p T\right) = \nabla \cdot (\lambda \nabla T) + Q_{losses} + S_{latent} + Q_{laser} \tag{6}$$

with

$$\begin{aligned} Q_{losses} = &\left[-\left(0.82 L_v \frac{M P_v}{\sqrt{2\pi M R T}}\right) - \sigma_b \epsilon \left(T^4 - T_{\text{ref}}^4\right) - h\left(T - T_{\text{ref}}\right)\right] \cdot \\ &|\nabla \gamma| \frac{2\rho C_p}{\rho_{\text{metal}} C_{p,\text{metal}} + \rho_{\text{gas}} C_{p,\text{gas}}}\,. \end{aligned} \tag{7}$$

and

$$S_{latent} = \gamma L \frac{\partial}{\partial t}\left(\rho f_l\right)\,. \tag{8}$$

In this equation, C_p, λ, σ_b, ϵ, h, L, and f_l denote the specific heat capacity, thermal conductivity, Stefan–Boltzmann constant, emissivity, heat transfer coefficient, latent heat of fusion, and liquid fraction, respectively. Eq. (6) accounts for transient heat transport through convection and diffusion, while incorporating surface heat losses, the latent heat of fusion/solidification, and a source term representing the laser energy input. As the latent heat term (Eq. (8)) depends on the temporal evolution of the liquid fraction, Eq. (6) is solved iteratively within each time step. This is coupled with an explicit source-based method for the solidification phase change, formulated as follows [28]:

$$\left[f_l\right]_{n+1} = \left[f_l\right]_n + \frac{1}{\left(T_l - T_s\right)}\left([T]_{n+1} - \left(\left[f_l\right]_n \left(T_l - T_s\right) + T_s\right)\right)\,. \tag{9}$$

Here, T_l and T_s denote the liquidus and solidus temperatures of the metal, respectively. The subscripts $n+1$ and n represent the values of the current and previous iterations, ensuring convergence within the iterative scheme.

The source term Q_{laser} in Eq. (6) is implemented via the ray-tracing laser class within the OpenFOAM framework. This approach discretizes the circular laser source into a finite number of individual rays. The power assigned to each discrete ray (i) is defined as follows:

$$P_i = \frac{2P}{\pi r_e^2} \exp\left[-2\frac{\left(x_i - vt - x_0\right)^2 + \left(z_i - z_0\right)^2}{r_e^2}\right]. \tag{10}$$

In this expression, P, r_e, v, x_i, x_0, z_i, and z_0 denote the total laser power, the effective beam radius, the scanning speed, the x-coordinate of the ith ray, the initial x-position of the laser, and the z- and y-coordinates of the starting point, respectively. Upon impact with a cell surface containing the metallic phase, a portion of the ray's energy is absorbed, while the remainder is reflected. The absorbed energy fraction is determined using the Fresnel absorption model

$$\xi_A = 1 - \frac{1}{2}\left(\frac{1 + (1 - \varepsilon\cos\theta)^2}{1 + (1 + \varepsilon\cos\theta)^2} + \frac{\varepsilon^2 - 2\varepsilon\cos\theta + 2\cos^2\theta}{\varepsilon^2 + 2\varepsilon\cos\theta + 2\cos^2\theta}\right) \tag{11}$$

which accounts for the local angle of incidence θ between the ray and the surface normal vector and the Fresnel coefficient ε [29–31]. For liquid stainless steel near the melting point, the absorptivity ξ_A at a normal incidence (0°) is approximately 30%, implying that 70% of the incident energy is reflected [32]. According to Mahrle et al. [33], the optical penetration depth of a fiber laser ($\lambda = 1074$ nm) in liquid iron at 1800 K can be estimated at 21.5 nm. Consequently, transmission effects can be neglected for cell sizes in the micrometer range, as the energy transfer occurs effectively at the surface.

2.3. Species conservation approach

The transport of nitrogen within the metallic phase is governed by the continuous species conservation equation, defined as follows:

$$\frac{\partial Y}{\partial t} + \nabla \cdot (\boldsymbol{u} Y) - \nabla \cdot \left(D_{N,l} \nabla Y\right) = \dot{Y}_{loss}\,, \tag{12}$$

where Y is the nitrogen mass fraction, $D_{N,l}$ the diffusion coefficient of nitrogen inside the steel and $\dot{Y}_{loss}$ the sink term at the free surface of the steel given as

$$\dot{Y}_{loss} = |\nabla\gamma| k (Y_{max,l} - Y) . \tag{13}$$

Eq. (12) accounts for transient nitrogen transport accounting for convection and diffusion. To guarantee a mass-conservative species transport, the advective term is coupled to a corrected, divergence-free phase flux. This modified flux is derived by solving a Poisson equation to satisfy the local continuity of the phase fraction, thereby suppressing any mass loss of the species across the interface into the gas domain. The relative importance of convective to diffusive mass transport is characterized by the Péclet number (Pe), defined as

$$Pe = \frac{|\boldsymbol{u}| \cdot 2r_l}{D_{N,l}} . \tag{14}$$

The sink term in Eq. (13) represents the degassing of nitrogen at the free surface of the melt pool. The driving force for this super-saturated degassing is the difference between the maximum solubility limit, $Y_{max,l}$, of nitrogen within the liquid steel and the local nitrogen concentration. This linear formulation assumes that the nitrogen outgassing process is primarily mass-transport limited. Given the extremely high surface temperatures during laser melting, the interfacial chemical reaction kinetics, specifically the recombination of dissolved nitrogen atoms into N_2 molecules, are assumed to proceed substantially faster than the mass transport of nitrogen from the bulk liquid to the interface. This justifies neglecting the interfacial reaction kinetics and employing a controlled mass transfer, first-order approximation driven by supersaturation. Furthermore, the nitrogen loss is governed by the mass transfer coefficient at the liquid-gas interface:

$$k = \frac{Sh \cdot D_{N,l}}{2r_l} \tag{15}$$

where $D_{N,l}$ is the diffusion coefficient and r_l denotes the focal laser radius. The Sherwood number Sh can generally be determined via empirical correlations involving the Reynolds (Re) and Schmidt (Sc) numbers. However, because the characteristic length in this framework is approximated by the focal laser diameter rather than derived from the hydrodynamic boundary layer, Sh is explicitly defined as an empirical Sherwood number in this study. It serves as an empirical calibration parameter and is assumed to be constant. The nitrogen loss is assumed to occur exclusively at the free surface, thereby neglecting gas bubble nucleation within the melt pool. This assumption is supported by experimental observations, as cross-sectional images of single-tracks without powder do not reveal any significant porosity within the melt pool. However, it must be noted that the absence of retained porosity does not exclude transient bubble formation and their subsequent escape before solidification. If such internal bubble dynamics occur, they would act as an additional nitrogen sink. Consequently, by attributing all mass loss exclusively to the free surface, the current model might overestimate the local surface degassing rates to compensate for any unmodeled internal depletion.

2.4. Physical properties and numerical setup

This section details the thermophysical material properties of the multi-phase system and the numerical configuration of the PBF-LB/M solver. The model is based on the VoF method, extended to incorporate phase transitions, species transport, and laser-matter interaction. The simulation domain comprises a multi-phase system consisting of the metallic alloy ($\gamma = 1$) and the surrounding gaseous atmosphere ($\gamma = 0$). Within the metallic phase, the solid (s) and liquid (l) states are distinguished based on the local temperature, T. Consequently, any thermophysical property ϕ within the metallic phase such as thermal conductivity or specific heat is determined as a weighted average of the solid and liquid properties:

$$\phi_{metal} = \phi_s \cdot (1 - l_f) + \phi_l \cdot l_f . \tag{16}$$

Table 1
Thermophysical properties for the PBF-LB/M simulation [34–39].

Property	Symbol	Value/Unit
Density	ρ_{metal}	$6814.48\,\mathrm{kg\,m^{-3}}$
Density	ρ_{gas}	$1.251\,\mathrm{kg\,m^{-3}}$
Viscosity	μ_{metal}	$10^{-3} e^{(-2.396+7950/T)}$ Pa s
Viscosity	μ_{gas}	1.758×10^{-5} Pa s
Thermal expansion	β_{metal}	$5.85 \times 10^{-5}\,\mathrm{K^{-1}}$
Thermal expansion	β_{gas}	$3.4 \times 10^{-3}\,\mathrm{K^{-1}}$
Specific heat	$C_{p,s}$	$470.973 + 0.15T$ J/(kg · K)
Specific heat	$C_{p,l}$	825 J/(kg · K)
Specific heat	$C_{p,gas}$	1009 J/(kg · K)
Thermal conductivity	λ_s	$20.143 + 5.655 \cdot 10^{-3}T + 1.617 \cdot 10^{-6}T^2$ W/(m · K)
Thermal conductivity	λ_l	$5.314 + 0.0139T$ W/(m · K)
Thermal conductivity	λ_{gas}	0.026 W/(m · K)
Solidus temperature	T_s	1673 K
Liquidus temperature	T_l	1750 K
Evaporation temperature	T_v	3096 K
Latent heat of fusion	L	$2.683 \times 10^5\,\mathrm{J\,kg^{-1}}$
Evaporation enthalpy	L_v	$6.295 \times 10^6\,\mathrm{J\,kg^{-1}}$
Surface tension	σ	$1.48 - 0.82 \cdot 10^{-3}(T - T_l)$ N m^{-1}
Heat transfer coefficient	h	$10\,\mathrm{W/(m^3 K)}$
Emissivity	ϵ	0.1
Reference temperature	T_{ref}	300 K
Fresnel coefficient	ε	0.18
Effective laser beam radius	r_e	23.4 µm
Focal laser beam radius	r_l	32.7 µm
Molar mass	M	$0.0546\,\mathrm{kg\,mol^{-1}}$
Carman-Kozeny constant	C_K	$1.5 \times 10^6\,\mathrm{kg\,m^{-3}\,s^{-1}}$
Compression coefficient	C_γ	1.0

The global property for a computational cell, considering both the metal and gas phases, is then determined using the VoF phase fraction

$$\phi_{cell} = \gamma_1 \phi_{metal} + (1 - \gamma_1)\phi_{gas} \quad \text{where } \phi \in \{\rho, \mu, \beta, C_p, \lambda\} . \tag{17}$$

Simulations and experimental trials were performed for the processing of Fe-15Cr-0.3C-N via PBF-LB/M under an N_2 atmosphere. Table 1 summarizes the relevant thermophysical properties utilized in the simulations for an alloy composed of 84.7 wt-% iron, 0.3 wt-% carbon, and 15 wt-% chromium.

The effective and focal laser beam radii (r_e, r_l) are direct experimental setup parameters. The characteristic phase transition temperatures (T_s, T_l) for the specific alloy composition (Fe-15Cr-0.3C-N) were determined via thermodynamic equilibrium calculations using the Thermo-Calc software package (databases TCFe10 and MOBFe5). In contrast, parameters such as L, L_v and T_v were calculated based on the mass fraction composition of the pure elements (Fe and Cr) [40]. Fundamental thermophysical properties such as ρ, β, C_p, λ were derived from established literature for the Fe-15Cr-0.3C-N system and for pure molecular nitrogen gas [34–39]. Due to the lack of high-temperature experimental data for this specific alloy, ϵ, ε, μ were adopted from an analogous, characterized stainless steel, 316L [25]. The surface tension σ and the Marangoni coefficient $d\sigma/dT$ were characterized for free-falling oscillating molten droplets of X45Cr13, utilizing an in-house on-demand droplet generator [41]. Furthermore, h was set to a standard value of 10 W/(m^2 K), a typical assumption for convective boundary losses above melt pools in LPBF simulations [19,25]. The diffusion coefficient $D_{N,l}$ and the maximum solubility of nitrogen within the melt pool $Y_{max,l}$ were derived and fitted for an atmospheric pressure of $p = 1.013 \times 10^5$ Pa. Because both the PBF-LB/M experiments and the numerical simulations were conducted under a 100% pure nitrogen process atmosphere, this total pressure corresponds to the nitrogen partial pressure. Consequently, if simulations were to be conducted under a mixed gas environment such as an argon atmosphere the thermodynamic calculations for the maximum solubility would need to be explicitly adjusted to account for the specific nitrogen partial pressure. Fig. 1 illustrates the temperature dependence of both

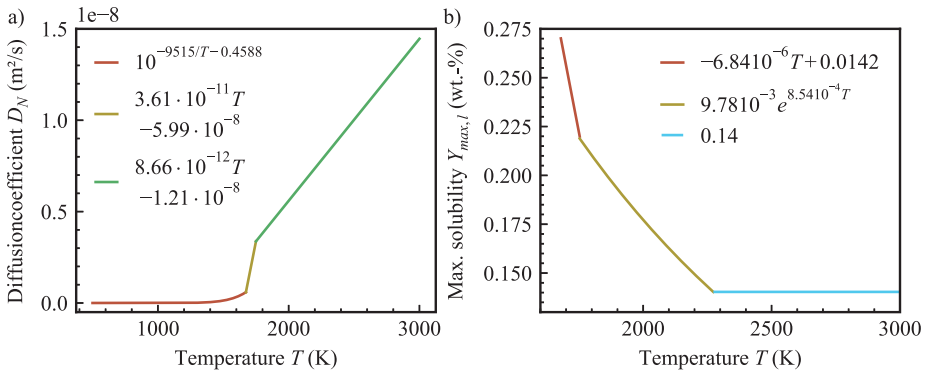


Fig. 1. (a) Diffusion coefficient and (b) maximum solubility of nitrogen inside the liquid steel melt (Fe-15Cr-0.3C-N) calculated with Thermo-Calc.

thermophysical properties for a pressure of $p = 1.013 \times 10^5$ Pa. For temperatures below the solidus ($T < 1673$ K), $D_{N,l}$ is derived under the assumption of a fully austenitic matrix, as nitrogen significantly stabilizes the austenite phase [42]. This assumption can be justified by the microstructural analysis of the solidified tracks, which confirms a predominantly austenitic grain structure at room temperature, as will be demonstrated in detail in Section 4.1. Beyond the threshold of 2273 K, $Y_{max,l}$ is assumed to remain constant at 0.14 wt-%. To systematically quantify the overall nitrogen loss across different processing conditions, the Nitrogen Retention Index (NRI) is introduced:

$$NRI = 1 - \left(\frac{Y_0 - \overline{Y}}{Y_0 - Y_{max,l}} \right) \tag{18}$$

This index describes the relative nitrogen retention within the processed material. Here, the mean nitrogen concentration $\overline{Y}$ is averaged over the entire volume that was in a liquid state at least once during the simulation, given that nitrogen transport and surface outgassing occur predominately within the melt pool. The fractional term in Eq. (18) compares the actual averaged concentration loss $Y_0 - \overline{Y}$ to the theoretically maximum possible loss $Y_0 - Y_{max,l}$. This baseline for maximum loss is defined by the initial nitrogen content of the substrate Y_0 and the aforementioned constant thermodynamic solubility limit at high temperatures $Y_{max,l} = 0.14$ wt-%. Consequently, the NRI is a dimensionless metric, which is expressed as a percentage in this study. A value of 100% indicates complete retention of the initial nitrogen content, whereas a value of 0% defines that the entire melted volume has completely lost the nitrogen down to its physical thermodynamic saturation limit. It should be acknowledged that alternative normalization strategies for the denominator such as using locally temperature-dependent solubility limits or different volume-averaging techniques are theoretically possible. However, the selected definition of the NRI represents a process-specific metric. Its primary purpose is to establish a standardized baseline that facilitates a robust comparative analysis of relative nitrogen loss across different PBF-LB/M processing parameters, rather than to serve as an absolute thermodynamic measure.

A critical aspect of the simulation model is the selective implementation of temperature-dependent properties. While some parameters are modeled as constants to maintain computational efficiency where temperature variations expected to be minimally influence the final result, temperature-dependent functions were explicitly implemented for the most critical variables. Specifically, μ_{metal}, Cp_s, λ_l, λ_s, σ, $D_{N,l}$ and $Y_{max,l}$ are defined as functions of the local cell temperature. Modeling the temperature dependence of surface tension and viscosity is strictly necessary, as these are the primary thermophysical drivers governing melt pool flow dynamics, keyhole formation, and the subsequent nitrogen outgassing. The specific impact of these sensitive parameters on the nitrogen loss is discussed in further detail in Section 4.4. Given the thermal gradients and rapid cooling rates inherent to the PBF-LB/M process, a brief estimation of data uncertainty is necessary. While the thermodynamic predictions (Thermo-Calc) offer a accurate baseline, they assume thermal equilibrium, which deviates from the non-equilibrium nature of laser melting, potentially shifting the effective transition temperatures slightly. The utilization of a linear mixture rule for evaporation properties neglects complex intermetallic interactions, leading to an estimated uncertainty range of 5% to 10%. Furthermore, adopting optical properties and viscosity from the 316L proxy introduces deviations, particularly near the boiling point, where experimental validation is challenging. Constants like ϵ are also temporally dynamic in reality, as they are strongly influenced by in-situ surface roughness and local oxidation. Despite these uncertainties, this approach combined with the selective temperature dependencies, provides a framework for capturing the phenomena observed in the experiments.

Single-track simulations were performed on a computational grid featuring an equidistant approach in the region of interest, with a cell size of 3.00 µm, a track length of 2000 µm, and a width of 290 µm. The cell size of 3.00 µm was selected based on a comprehensive grid independence study. To ensure that the numerical smearing of the VoF interface does not affect the calculated nitrogen loss, the grid sensitivity was evaluated for the nitrogen depth profiles using resolutions of 3.00 µm, 3.78 µm, and 4.76 µm (see Fig. A.11). The results confirm that the chosen resolution provides a grid-independent solution for the interfacial mass transfer. In the peripheral regions, a non-uniform mesh refinement was applied to optimize computational efficiency, resulting in a total of approximately 9.3 million cells. All governing equations were resolved within the OpenFOAM v2112 framework using a modified interFoam solver. The pressure-velocity coupling was managed via the PIMPLE algorithm, which enhances numerical stability when coupled with an adaptive time-stepping scheme. The time steps were dynamically adjusted, constrained by a maximum duration of 0.5 µs and a maximum Courant number of $Co = 0.7$. To establish the numerical framework for the PBF-LB/M process, a comprehensive set of initial and boundary conditions was implemented. The domain was initialized as a multi-phase system, where the metallic phase and the N_2 atmosphere were distinguished by the phase fraction field γ. The initial thermal state was set to a uniform temperature of 300 K, and the initial pressure was defined as 1.013×10^5 Pa. Fluid dynamics were governed by noSlip conditions at the solid boundaries and pressureInletOutletVelocity at the atmospheric boundaries to facilitate gas transport. For the pressure field p, fixedFluxPressure was applied at the walls, while a totalPressure of 0 Pa was prescribed at open boundaries. Thermal transport at the domain limits was modeled using the externalWallHeatFluxTemperature condition, with a heat transfer coefficient of 10 W/(m²K) and phase-specific thermal conductivities of 0.026 W/(mK) for the gas phase and 22.36 W/(mK) for the metallic walls. Other scalar fields, including the nitrogen mass fraction Y and the liquid fraction f_l, were assigned zeroGradient conditions at all domain boundaries.

All simulations were performed on a high-performance computing (HPC) cluster utilizing two nodes, each equipped with 2x Intel Xeon

Table 2
Laser processing parameters and calculated surface energy densities.

Track (–)	Laser power P (W)	Scanning speed v (mm/s)	Energy density E_A (J/mm²)
1	150	900	2.55
2	150	990	2.32
3	150	1080	2.12
4	150	1170	1.93

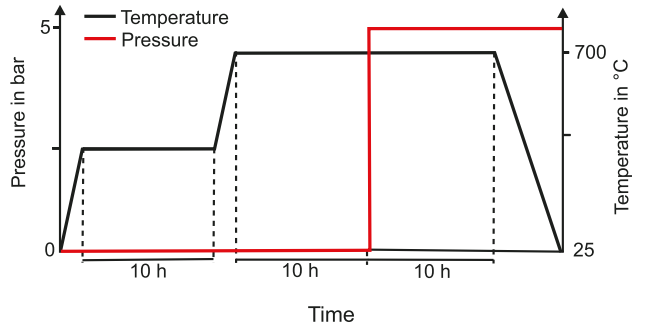


Fig. 2. Temperature and pressure profile of the applied nitriding process.

Platinum 9242 processors, totaling 192 cores. The single-track simulations were conducted at a constant laser power P of 150 W. First, a sensitivity analysis was conducted for three different Sh numbers (Sh = 200, 300, and 400) at a fixed scanning speed of 900 mm s^{-1} and an initial nitrogen content of 0.75 wt – %, corresponding to the concentration measured via WDS within the stainless steel substrates used in the experiments. This step served to calibrate and validate the numerical model by comparing the magnitude and the trend of the resulting nitrogen concentration profiles at the surface with experimental data. Subsequently, using a constant Sh number of 300, further simulations were carried out across four varying scanning speeds to investigate the influence of the surface energy density, defined as $E_A = \frac{P}{2r_l v}$ (see Table 2).

The numerical simulations in this parameter study were performed using a nitrogen concentration of Y_0 = 0.71 wt – %. While subsequent experimental analysis suggested a slightly higher concentration of Y_0 = 0.75 wt – %, the simulation results remain highly representative. The primary objective was to capture the fundamental transport phenomena and concentration profiles, which show low sensitivity to minor variations in concentration within this regime.

3. Experiments - materials and methods

3.1. Production of substrate materials

Single-track experiments were performed on two stainless steel substrates (Fe-15Cr-0.3C-N) with distinct initial nitrogen concentrations: a low nitrogen substrate containing 0.11 wt-%, below the maximum melt solubility limit, and a high nitrogen steel substrate with 0.75 wt-% measured with WDS. The substrates were produced by hot isostatically pressing gas atomized powder. Gas atomization was performed with N_2 as the melting atmosphere and atomization medium. The atomized powder was subsequently gas nitrided in a pressure-assisted heat treatment in a vacuum furnace. Nitriding was conducted in an N_2 atmosphere following the temperature and pressure profile illustrated in Fig. 2. Table 3 summarizes the chemical composition of the low- and high nitrogen powder, measured via Optical Emission Spectrometry and Carrier Gas Hot Extraction for N and O. The N-content of 0.76 wt-% was achieved by mixing low nitrogen powder with nitrided powder.

Consolidation was performed in a HIP QIH9 from Quintus Technologies for 180 min at a temperature of 1150 °C and pressure of

Table 3
Chemical composition of initial and mixed Fe-15Cr-0.3C-N powder in wt-% and ppm for O.

	C	Si	Mn	P&S	Cr	Ni	Mo	N	O	Fe
Gas-atomized	0.26	0.49	0.67	0.02	15.22	0.02	0.02	0.1108 ± 0.0005	182 ± 5	bal.
Nitrided & mixed	”	”	”	”	”	”	”	0.764 ± 0.013	238 ± 1	bal.

105 MPa with subsequent solution annealing at 1100 °C for 180 min with oil quenching in a furnace under air atmosphere. The capsule was then removed by turning and separated into discs with wire electrical discharge machining, which were finally sandblasted.

3.2. PBF-LB/M single-tracks and microstructural investigations

On each substrate, four single-tracks were produced using the laser processing parameters specified in Table 2. This experimental setup was designed to investigate the combined influence of energy density and initial nitrogen content on the resulting melt pool dimensions. The material was processed using an Aconity-MINI system equipped with a single-mode fiber laser (wavelength λ = 1070 nm, nominal power P_{max} = 400 W, and effective beam radius r_e = 23.4 µm). After PBF-LB/M processing, the specimens were sectioned, ground with 500 to 1000 mesh and polished down to 1 µm using diamond suspension. To facilitate metallographic analysis samples were etched using a V2 A etchant (100 ml distilled water, 100 ml 36% hydrochloric acid, 10 ml nitric acid). Each single-track was evaluated across three distinct cross-sections using optical microscopy.

3.3. Measuring of local nitrogen contents

In addition to optical microscopy, track 1 of the high nitrogen series was characterized using a scanning electron microscope (SEM) MIRA3 from Tescan equipped with WDS to determine the nitrogen concentration profiles across the vertical center axis of melt pool cross sections. For WDS, a WAVE detector (Oxford Instruments) and an LSM60 crystal were used, with the electron gun operated at 10 kV. The working distance was 15 mm. For nitrogen standardization, X30MnCrN19-18 (CN 0.96, [43]) with a nitrogen content of 0.61 wt-% was employed. The WDS measurements were performed across two distinct cross-sections to ensure statistical relevance. The resulting experimental concentration profiles serve as the basis for calibrating the mass transfer coefficient by varying the Sh number as detailed in Section 4.2.

4. Results and discussion

4.1. Influence of nitrogen on microstructural evolution

The cross-sectional micrographs in Fig. 3 illustrate the pronounced influence of the nitrogen content on the solidification morphology and phase evolution of Fe-15Cr-0.3C-N steel tracks produced via Laser Powder Bed Fusion (PBF-LB/M). In the low nitrogen specimen (see Fig. 3a), the melt pool as well as the substrate exhibit a predominantly homogeneous, fine-scaled internal structure. This morphology is indicative of martensitic transformation, as the rapid cooling rates inherent to the PBF-LB/M process facilitate the formation of martensite within the prior austenite grains. In contrast, the high nitrogen cross-section (see Fig. 3b) reveals a distinct solidification pattern characterized by cellular-dendritic growth. Primary dendritic arms extend epitaxially from the melt pool boundary toward the center, aligned with the direction of the maximum thermal gradient. Since nitrogen acts as an austenite stabilizer, it significantly depresses the martensite start

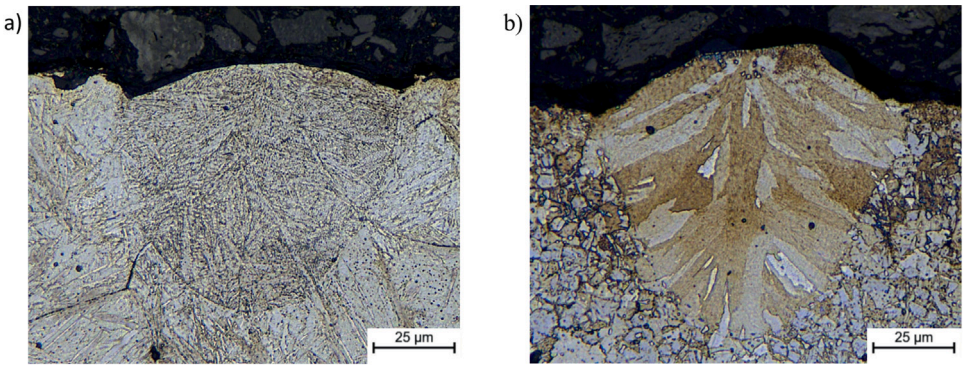


Fig. 3. Single-track cross sections for Fe-15Cr-0.3C-N with (a) low nitrogen content (0.11 wt-%) and (b) high nitrogen content (0.75 wt-%) printed with P = 150 W and v = 900 mm/s.

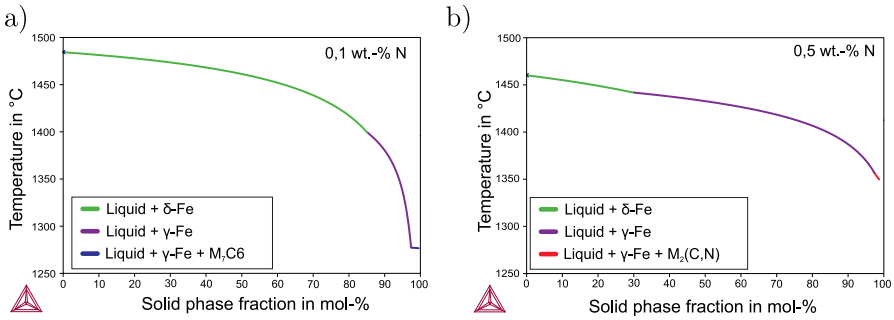


Fig. 4. Solidification sequence of steel Fe-15Cr-0.3C-N calculated using solidification simulations based on the Scheil–Gulliver method in Thermo-Calc®.

temperature [12]. This is also reflected in the primary solidification sequences calculated using Scheil–Gulliver. Fig. 4a presents the simulation for Fe-15Cr-0.3C-N with 0.1 wt-% nitrogen and Fig. 4b for 0.5 wt-% N. The simulations indicate that both materials undergo primary ferritic solidification. Comparison of the two simulations clearly shows that a higher nitrogen content shifts the transition to an austenitic solidification sequence to higher temperatures, i.e., to an earlier stage of solidification. For the low nitrogen material, the simulation indicates primarily ferritic solidification of approximately 87 mol-%, followed by austenitic phase. The proportion of primarily ferritic solidification in the high nitrogen material is only about 30 mol-%. In contrast, the microstructure of the high nitrogen single-track shown in Fig. 3b does not exhibit a two-phase ferritic-austenitic structure. Instead, it is interpreted as primary austenite due to the columnar and epitaxial grain growth. While Scheil–Gulliver simulations indicate the austenite-stabilizing effect of nitrogen, they cannot be used to predict the resulting microstructure after PBF-LB/M, as the non-equilibrium conditions during the process are not taken into account. Among others, Vitek et al. [44] and David et al. [45] found that at high cooling rates, primary austenitic solidification is favored over ferritic solidification. They attribute this to more favorable kinetics of nucleation and growth of austenite grains.

Fig. 5 presents the measured melt pool widths at the substrate surface and the corresponding melt pool depths for tracks 1–4 (see Table 2), illustrated as a function of the initial nitrogen content. The comparison between low and high nitrogen contents reveals that the melt pools are slightly larger in the high nitrogen specimens. The width and depth of the melt pool are fundamentally governed by a dynamic force balance: the recoil pressure acts to open and deepen the depression zone, while the surface tension acts to close it. In liquid steel, nitrogen acts as a surface-active element that slightly lowers the surface tension [46]. Furthermore, it is hypothesized that the intensified surface degassing observed at higher nitrogen concentrations might also induce a locally increased recoil pressure. Consequently, the combination of a nitrogen-induced reduction in surface tension combined with a potentially enhanced recoil pressure shifts the interfacial force balance, allowing for deeper depression zone and ultimately resulting in a larger overall melt pool volume.

4.2. Calibration and validation with nitrogen profiles

The longitudinal view in Fig. 6a illustrates the final nitrogen distribution, characterized by minimum nitrogen concentrations at the melt track surface and a progressive increase toward the substrate. Furthermore, reaching a fully developed melt pool during single-track formation requires an initial period, as the energy is primarily consumed by surface melting. This results in depleted nitrogen levels at the leading edge (left) of the melt track's surface region. At the trailing edge (right) of the single-track, the laser leaves behind a solidified depression zone, establishing a stable region between the entry and exit points of the track. The nitrogen concentrations of all computational cells within the longitudinal center plane ($z = 0$) of this stable region (see Fig. 6c) are extracted and longitudinally averaged along the scanning direction (x-axis) to determine the nitrogen concentration profiles relative to the melt pool depth (indicated by the black arrow along the vertical y-axis in Fig. 6b). Subsequently, these averaged nitrogen profiles are plotted against the absolute melt pool depth for three distinct Sh numbers and compared with experimental WDS line scans (see Fig. 6d). The numerical results exhibit significant nitrogen gradients near the surface (Zone I), followed by a nearly linear increase throughout the melt pool depth (Zone II), which ends with an almost discrete jump to the initial substrate concentration (Zone III). The parameter study reveals that higher Sh numbers shift the nitrogen profiles toward lower concentrations. This relation is attributed to increased mass transfer coefficients that enhance the nitrogen desorption rate (see Eqs. (13) and (15)). The initial experimental data point is located at a depth of 10 µm, a distance intentionally selected to avoid interference from the embedding material. The nitrogen concentration at this point is slightly higher than the numerical predictions for the given Sh numbers

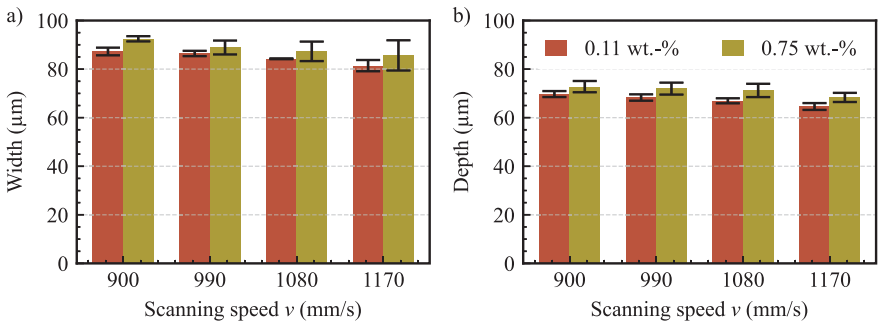


Fig. 5. Experimentally measured (a) melt pool width and (b) melt pool depth for Fe-15Cr-0.3C-N with low nitrogen content (0.11 wt-%) and high nitrogen content (0.75 wt-%).

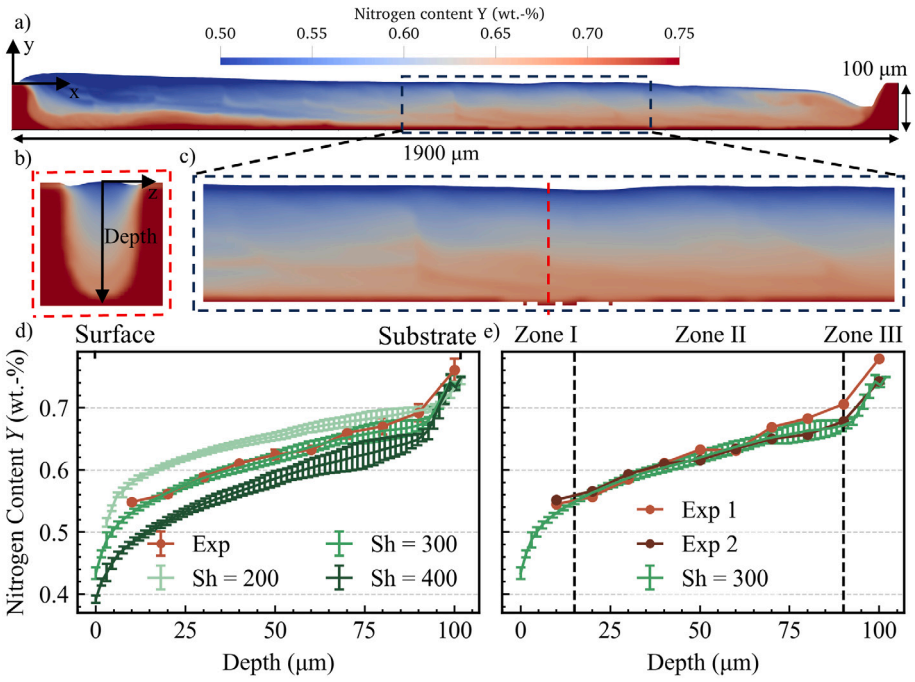


Fig. 6. Final nitrogen content for Fe-15Cr-0.3C-N with $P = 150\,\mathrm{W}$, $v = 900\,\mathrm{mm\,s^{-1}}$ and $Y_0 = 0.75\,\mathrm{wt-\%}$ (a) in longitudinal view (b) in cross-sectional view at the red dotted line and (c) inside stable region. Nitrogen concentration profiles (d) for $Sh = 200$, 300, 400 and for the averaged WDS line scan and (e) and the reproducibility of the concentration profile.

and does not reflect the steep gradients observed in the simulations. This discrepancy may be attributed to the neglection of solid-state nitrogen absorption during the cooling phase. Nevertheless, the averaged nitrogen profile for Sh = 300 shows good agreement with the experimentally measured profile. The Sh number also directly affects the global nitrogen loss, which is quantified by the NRI. A sensitivity analysis reveals that varying the Sh number from $Sh = 200$ to $Sh = 400$ results in a corresponding shift of the overall NRI from 87.0% to 76.8%. This significant dependence highlights that the Sh number strongly controls the absolute magnitude of the outgassing kinetics. It must be clarified that while the Sh number is calibrated to match the absolute magnitude of the nitrogen loss, the underlying physical transport model is independently validated. Specifically, the model successfully predicts the distinct slopes and concentration gradients, which are governed by the melt pool fluid dynamics rather than the chosen Sh value. Fig. 6e compares both measured WDS line scans with the numerical nitrogen profile. Both experimental profiles are in good agreement, although Experiment 1 exhibits a slightly higher nitrogen concentration in the substrate, leading to minor variations in the averaged comparison (see Fig. 6d). Experiment 2 shows a nearly perfect fit to the model (Zone II and Zone III) and exhibits gradients similar to Experiment 1.

4.3. Nitrogen degassing process

Having calibrated and validated the single-track simulation, the next objective is to analyze the mechanisms of nitrogen transport and depletion within the melt pool. Consequently, Fig. 7 and Supplementary Video 1 illustrate the temporal evolution of nitrogen concentration as observed in the longitudinal center plane of the simulation domain.

The snapshot at $t = 0.2$ ms illustrates the initial formation of the keyhole, which achieves a fully developed, stable morphology by $t = 0.94$ ms. Simultaneously, the melt pool boundaries show a fully developed melt pool at $t = 0.94$ ms and $t = 1.68$ ms, where the melt pool exhibits a significant longitudinal extension behind the keyhole, a phenomenon attributed to the low thermal diffusivity of the steel compared to other metals [34,36]. To provide a more comprehensive analysis, Fig. 8 depicts the distribution of relevant physical quantities within the longitudinal center within the melt pool at $t = 1.68$ ms, corresponding to the stable region (see Fig. 6c). The temperature distribution within the melt pool reveals the highest temperatures inside the keyhole at the surface, where energy absorption is intensified by the Gaussian beam profile and multiple laser reflections.

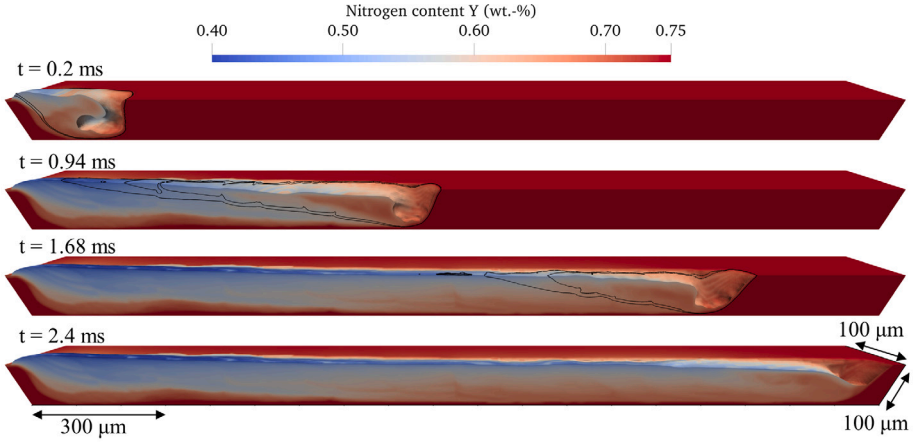


Fig. 7. Nitrogen contents in the longitudinal view for Y_0 = 0.75 wt-%, P = 150 W, v = 900 mm/s and Sh = 300 at different time steps together with solidus and liquidus temperature boundaries (black line).

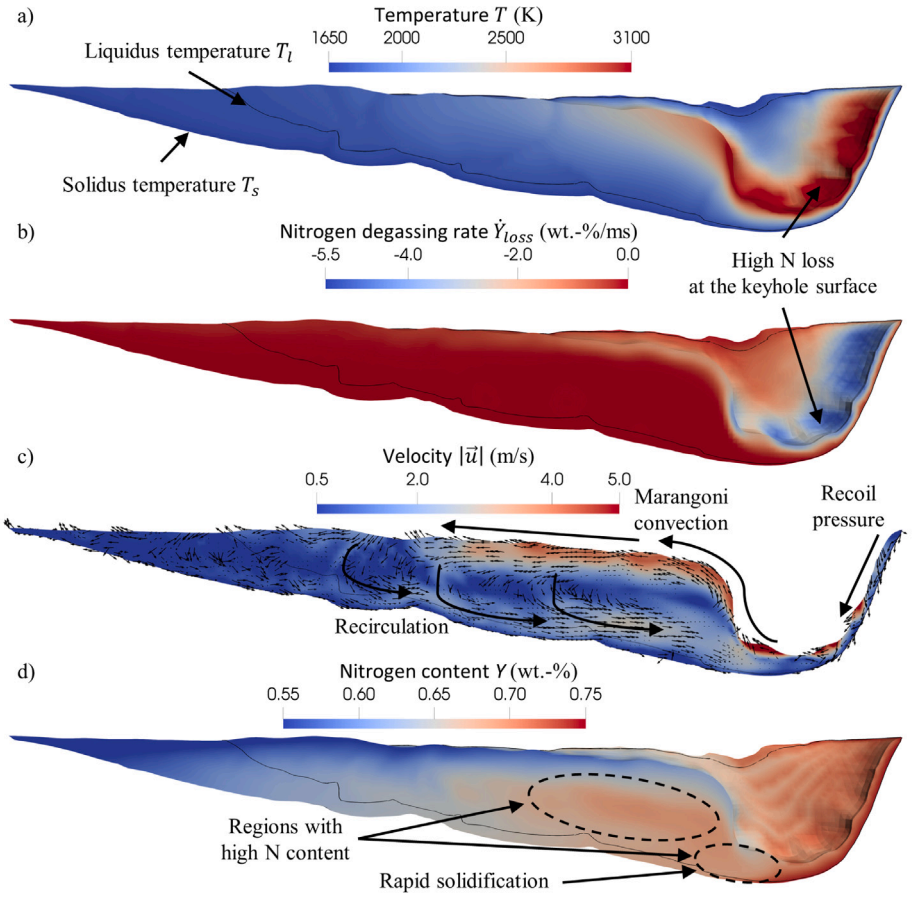


Fig. 8. Longitudinal view of the melt pool for Y_0 = 0.75 wt-%, P = 150 W, v = 900 mm/s and Sh = 300 at t = 1.68 ms showing the (a) temperature distribution (b) the local nitrogen desorption rate (c) the magnitude of the velocity in combination with uniformly sized vectors and (d) the nitrogen content.

The local degassing rate (see Fig. 8b) exhibits the most significant nitrogen loss in these high-temperature regions, reaching rates of −5.5 wt-%/ms, as mass transfer scales directly with temperature. Specifically, elevated temperatures enhance nitrogen diffusion while simultaneously reducing the maximum nitrogen solubility. This combination increases both the mass transfer coefficients and the driving forces for degassing. Furthermore, nitrogen outgassing persists at the surface behind the keyhole at rates of −2.0 wt-%/ms. Fig. 8c and Supplementary Video 2 illustrate the velocity magnitude alongside vectors indicating the direction of momentum. At the leading edge of the melt pool, recoil pressure drives momentum downward into the keyhole. Conversely, at the trailing edge, Marangoni forces transport heat and N-depleted steel melt (see Fig. 8d and Supplementary Video 3) toward the substrate surface. As illustrated by the spatial distribution in the Appendix (see Fig. B.12), the local Pe number within the melt pool ranges from 500 to 50 000, conclusively demonstrating that mass transport is predominantly driven by convection ($Pe \gg 1$). These transport phenomena, in conjunction with persistent surface degassing,

result in the steep nitrogen gradients observed at the solidified surface in Zone I (see Fig. 6e). Another observed phenomenon is that the narrowing of the elongated melt pool depth behind the keyhole increases drag forces at the solid–liquid boundary. Consequently, momentum conservation leads to recirculation zones with high velocity magnitudes and Pe numbers. This results in a downward and forward-directed flow from the surface back toward the keyhole. Zhang et al. [47] reported similar recirculation zones for a single-track simulation on 316L stainless steel with an energy density of $E_A = 3.0\,\mathrm{J\,mm^{-2}}$, which is slightly higher than the $E_A = 2.55\,\mathrm{J\,mm^{-2}}$ used in the present case. He observed recirculation vortices in the center and rear of the melt pool, occurring as the momentum of the surface liquid flowing toward the rear is redirected downward and forward upon encountering the solidification front. Supplementary Video 3 demonstrates how these recirculation vortices transport nitrogen depleted melt deeper into the pool, thereby reducing the nitrogen concentration in Zone II. Fig. 8d illustrates a persistent region of higher nitrogen content in the melt pool center. However, as more N-depleted melt is pushed into this region toward the rear, a linear increasing nitrogen profile develops in Zone II (see Fig. 6e). Finally, rapid solidification in the deeper substrate regions suppresses these recirculations, leading to a concentration jump in Zone III up to the initial substrate level of $Y_0 = 0.75$ wt-%.

While a comprehensive sensitivity analysis falls outside the scope of this study, it is crucial to identify the most significant sources of uncertainty governing the predicted nitrogen transport. A primary numerical consideration is the representation of the liquid-gas interface within VoF framework. In this study, interfacial mass transfer is mathematically distributed across the transition region using the magnitude of the phase fraction gradient ($|\nabla\gamma|$). This approach implies that the calculated nitrogen loss exhibits a dependence on the thickness of the numerical interface that dictates the volume over which the sink term Y_{loss} is applied. Finally, the accuracy of the degassing model is highly sensitive to uncertainties on transport properties like heat diffusivity, heat loss at the free surface and kinematic viscosity. The solubility limit $Y_{max,l}$ and the diffusion coefficient $D_{N,l}$ are strictly temperature-dependent, uncertainties in the predicted surface temperature directly affect the thermodynamic driving force for outgassing and together with the kinematic viscosity the intensity of the Marangoni-driven convective transport.

4.4. Effect of surface energy density

Following the characterization of the nitrogen transport mechanisms at $900\,\mathrm{mm\,s^{-1}}$, the study is now extended to a broader range of process parameters to evaluate their impact on the nitrogen loss. Fig. 9 shows the longitudinal center plane for the parameter study (see Table 2) with different energy densities by varying the scanning speed. The primary distinction lies in the decreasing penetration depth of the keyhole, which leads to a reduction in overall melt pool depth and, consequently, diminished nitrogen loss in the deeper substrate regions. To validate the laser penetration, Table 4 provides a comparison between the measured and numerically extracted melt pool widths and depths referenced at the substrate surface. For melt tracks 1 and 2, the simulation overestimates the depth compared to the experimental results, whereas the opposite trend is observed for the width. Nevertheless, the simulated melt pool geometries show reasonable agreement with the experiments. A comparison of the melt pool cross-sections (see Figs. 3b and 6b) reveals that the experimental melt pool expands further across the substrate surface than the simulation. To provide a clearer assessment of the melt error of the melt pool prediction, the relative errors for width and depth have been included in 4. The data indicates a specific trend in some cases (e.g., Track 2), where the simulation slightly underpredicts the melt pool width while overpredicting its depth. This geometrical deviation can be attributed to approximations in the fluid flow driving forces and optical properties. Specifically, utilizing high-temperature properties derived from a

Table 4
Comparison of melt pool sizes and NRI.

Track (–)	v (mm/s)	Y_0 (wt-%)	Width			Depth			NRI (%)
			(μm)		(%)	(μm)		(%)	
			Exp.	Sim.	Err.	Exp.	Sim.	Err.	
1a	900	0.75	92.5	76.6	−17.2	72.7	92.7	+27.5	81.7
1b	900	0.71	92.5	76.6	−17.2	72.7	92.7	+27.5	81.6
2	990	0.71	88.9	71.7	−19.3	72.0	81.3	+12.9	82.6
3	1080	0.71	87.3	68.1	−22.0	71.2	72.1	+1.3	83.0
4	1170	0.71	85.6	66.0	−22.9	68.3	64.5	−5.6	83.3

316L such as the kinematic viscosity might underestimate the intensity of outward-directed Marangoni convection at the free surface, which would physically widen the melt pool. Furthermore, the simplified Fresnel absorption model inside the keyhole may input the absorbed laser energy slightly too efficiently into the vertical depth rather than laterally, leading to the observed deeper but narrower simulated melt pool geometries. However, when considering the absolute depth referenced to the solidified surface (see Fig. 6), the simulation and experiment show a higher degree of correlation. The nitrogen distributions within the melted volume appear qualitatively similar across the various energy densities. To quantify the nitrogen loss within the parameter study, the NRI results (see Eq. (18)) are presented in Table 4 and reveal that an increased energy density leads to intensified nitrogen loss. A comparison between Track 1a and 1b shows that the relative loss remains constant for different initial nitrogen contents. This implies that the absolute amount of degassed nitrogen increases with higher Y_0 due to the greater driving forces associated with super-saturation. To further evaluate the parameter study, nitrogen profiles are now plotted over the normalized melt pool depth (see Fig. 10a) using the same methodology as previously described (see Section 4.2). Normalization facilitates a more comprehensive comparison and reveals a consistent trend across varying energy densities. Specifically, nitrogen concentrations at the melt track surface decrease as energy density increases. Higher surface energy densities result in reduced cooling rates, thereby prolonging the duration of nitrogen degassing [48]. Conversely, an opposite trend is observed in Zone II, preceding the transition to Zone III. In this region, higher scanning speeds correlate with lower nitrogen concentrations. This phenomenon can be attributed back to the velocity fields within the melt pool (see Supplementary Fig. C.13). At the trailing edge of the keyhole and the subsequent surface region, Marangoni convection exhibits comparable magnitudes across all four scanning speeds. Simultaneously, as the volume of the melt pool decreases, the solidification rate at the advancing solid–liquid interface increases for higher scanning speeds [49]. Consequently, to satisfy momentum conservation, the resulting reverse flow exhibits significantly higher velocity magnitudes (see Fig. C.13), which are further intensified by the rapidly moving solidification front. These intensified reverse flows at higher scanning speeds facilitate the convective transport of N-depleted melt from the surface into the deeper regions on the normalized scale of the melt pool. Fig. 10b displays the NRI relative to the normalized melt pool depth, revealing a self-similarity across the two varying initial nitrogen concentrations in the substrate. These findings are consistent with the NRI values calculated from the mean nitrogen contents for Tracks 1a and 1b (see Table 4). Theoretically, this observed self-similarity enables the analytical prediction of nitrogen profiles for varying initial nitrogen contents.

4.5. Limitations and scalability

A key limitation of the current species transport framework lies in the mathematical formulation of the mass transfer coefficient (see Eq. (15)). First, the simplification of assuming a constant empirical Sh number has to be discussed. Bird et al. [50] summarized Sh correlations

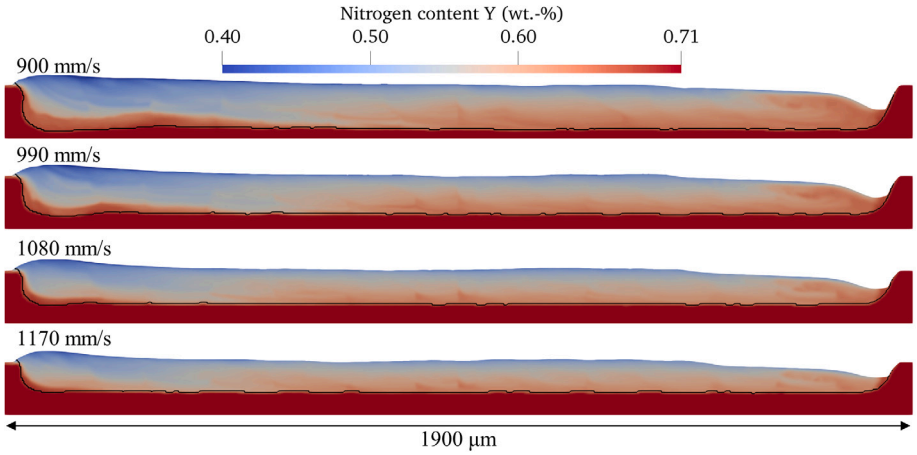


Fig. 9. Final nitrogen contents in the longitudinal center plane for Y_0 = 0.71 wt-%, P = 150 W and Sh = 300 for varying scanning speeds together with the solidification boundary (black line).

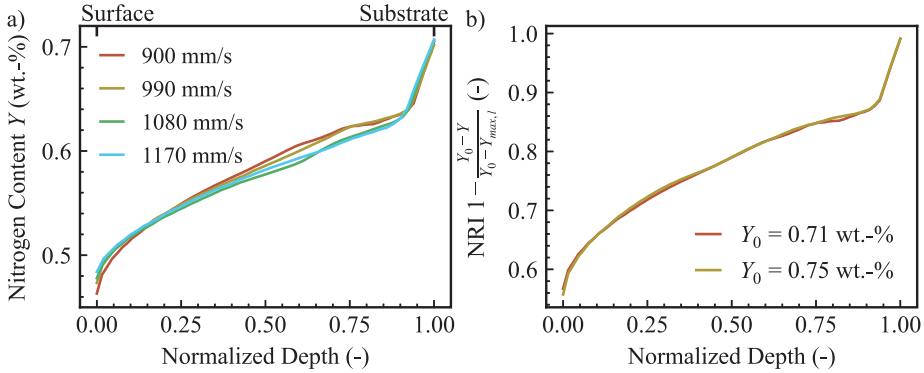


Fig. 10. Nitrogen concentration profiles and NRI over the normalized melt pool depth for (a) different scanning speeds and (b) different initial substrate concentrations.

based on interface geometries and properties. The Sh number scales with the Re- and Sc-number, both of which are highly temperature-dependent. Nevertheless, since the accurate calculation of a local Sh number within a PBF-LB/M induced melt pool remains exceptionally challenging due to complex interface shapes and dominant phenomena such as Marangoni convection and recoil pressure, a constant Sh approach was adopted for this study. This simplification is connected to the definition of the characteristic length scale. Currently, the focal laser diameter ($2r_l$) is utilized as a macroscopic approximation for the mass transfer zone. This introduces a physical inaccuracy, as this optical process parameter is only indirectly coupled to the actual hydrodynamic boundary layer governing interfacial mass transfer. To improve the physical fidelity, future iterations of the model must be extended to dynamically compute both the local Sh number and the characteristic length scale individually from the evolving flow field characteristics, rather than relying on an empirical Sherwood number. Ultimately, the primary objective of this formulation is to provide a reasonable estimation of the temperature-dependent mass transfer coefficient k that drives the super-saturated degassing kinetics. The current approach proves that setting Sh = 300 results in mass transfer coefficients ranging from $k(T_l = 1750\,\mathrm{K}) = 0.014\,\mathrm{m\,s^{-1}}$ to $k(T_v = 3096\,\mathrm{K}) = 0.067\,\mathrm{m\,s^{-1}}$, leading to a predicted nitrogen desorption that is in good agreement with the conducted experiments.

It is relevant to discuss further limitations of the numerical model employed in this study. Nitrogen transport is modeled as a one-way coupled passive scalar: the concentration field is determined by pre-calculated velocity and temperature distributions, while the back-coupling of local nitrogen content on thermophysical properties is neglected. This assumption is justified by the negligible variations in experimental melt pool dimensions observed across different initial nitrogen concentrations (see Fig. 5). However, for nitrogen contents exceeding Y_0 = 0.75 wt-%, the influence of back-coupling resulting from nitrogen transport and depletion may become more significant. The assumption of a one-way coupled species transport is further justified by a scale analysis of the interfacial shear stresses. As detailed in Appendix D, the temperature-driven Marangoni stress is more than an order of magnitude (approximately 12 times) stronger than the concentration-driven solutal Marangoni stress, making the solutal back-coupling effect on the macroscopic fluid dynamics negligible. Furthermore, variations in the Re and Sc numbers across wider parameter windows will impact the validity of the constant Sh number simplification, which directly affects the nitrogen degassing rate at the free surface. Thermodynamic equilibrium calculations indicate that the gas phase forms at temperatures exceeding 2300 K, suggesting that heterogeneous nucleation of molecular nitrogen may occur within the super-heated melt pool. This phenomenon could lead to additional nitrogen loss, beyond the depletion at the free surface of the melt pool, due to the growth of nitrogen bubbles inside the super-saturated high nitrogen steel.

Regarding scalability, the novel high-fidelity model with species transport for nitrogen can be extended to simulations with powder layers on top of the substrates by adjusting the initial volume fraction field. However, powder layers were intentionally neglected in this study to isolate the fundamental outgassing kinetics and to ensure a rigorous experimental validation, which becomes more complex for multi-layer builds.

5. Conclusion and outlook

In conclusion, the study successfully establishes a transient particle-scale high-fidelity model utilizing the VoF method in combination with conservative species transport to systematically evaluate nitrogen transport and degassing kinetics during the PBF-LB/M processing of high nitrogen martensitic stainless steel. By calibrating the magnitude and validating the trend of the nitrogen concentration profiles within the melt pool, it could be confirmed that the numerical model exhibits good agreement with experimental data concerning both the resulting melt pool dimensions and the quantified nitrogen loss. Furthermore, experimental measurements revealed that substrates with higher initial nitrogen contents produce slightly larger melt pools. This phenomenon is likely attributed to an increased recoil pressure resulting from the intensified degassing at the free surface. The final nitrogen concentration profile across the melt pool depth is primarily governed by intense degassing at the keyhole surface, coupled with convective transport mechanisms. Marangoni convection and recoil pressure initially drive N-depleted liquid toward the rear melt pool surface, inducing steep local concentration gradients. Subsequently, recirculation vortices transport this depleted melt into intermediate depths to establish an almost linearly increasing nitrogen profile, until the rapidly advancing solidification front suppresses these fluid flows at the melt pool bottom, resulting in an abrupt concentration jump back to the initial high nitrogen substrate level. The parametric study employs a newly introduced NRI to quantify relative nitrogen loss across different energy densities by varying scanning speeds. The results demonstrate that higher energy densities prolong active surface degassing, thereby intensifying the overall nitrogen loss. Conversely, higher scanning speeds induce stronger reverse fluid flows that convectively transport N-depleted melt into deeper regions on normalized scales of the melt pool. Notably, the NRI exhibits self-similarity across different initial nitrogen substrate concentrations.

In the future, the current numerical framework will be validated across a broader process window using further WDS measurements. Such comprehensive validation could potentially facilitate the derivation of an empirical correlation for the Sh number in PBF-LB/M induced melt pools, which was simplified as a constant in the present study but is expected to vary with complex fluid dynamics. The integration of more recent thermodynamic data bases, such as from Thermo-Calc, will allow for the direct calculation of temperature-dependent thermophysical properties tailored specifically to the investigated alloy composition, thereby further reducing the reliance on literature-based approximations. Furthermore, given that the actual PBF-LB/M process inherently utilizes a powder bed, future investigations must advance to incorporate single-, multi-track and multi-layer simulations that explicitly include powder layers, which were intentionally neglected in this study to isolate fundamental outgassing kinetics. Investigating multi-layer builds is particularly crucial, as the iterative remelting during the deposition of subsequent powder layers will influence the final nitrogen concentration profiles within the underlying layers. In this context, attention must be given to the penetration depth of the melt pool, which dictates the extent of this localized remelting and is governed by the applied energy density.

CRediT authorship contribution statement

Lennart Buchholz: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sophia Garthe:** Writing – original draft, Validation, Investigation, Data curation. **Louis Becker:** Writing – review & editing, Validation, Supervision, Investigation, Data curation. **Lutz Mädler:** Writing – review & editing, Supervision. **Sebastian Weber:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Nils Ellendt:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Gemini, DeepL, and NotebookLM in order to conduct literature research, summarize key findings from research papers, bring equations into the right formatting, and refine the scientific English phrasing. Furthermore, Gemini was utilized to assist in the C/C++ implementation of the numerical models and for the post-processing of data. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Grid convergence study

To validate the numerical robustness of the interfacial nitrogen sink term ($\dot{Y}_{loss}$), a grid independence study was conducted for the nitrogen transport model. Fig. A.11 illustrates the longitudinally averaged centerline nitrogen depth profiles for three different base cell sizes: 3.00 µm, 3.78 µm, and 4.76 µm. The corresponding averaged mean nitrogen contents are 0.606 wt-%, 0.610 wt-%, and 0.621 wt-%, respectively. The predicted nitrogen profiles show excellent spatial agreement across the entire depth of the melt pool. The maximum deviation in the global mean nitrogen content between the coarsest (4.76 µm) and the finest (3.00 µm) mesh is less than 2.5%. Consequently, the nitrogen concentration field is considered grid-independent, confirming that the chosen resolution of 3.00 µm sufficiently resolves the coupled momentum and mass transport phenomena without introducing significant numerical artifacts at the liquid-gas interface.

Appendix B. Péclet number

See Fig. B.12.

Appendix C. Velocity fields for varying scanning speed

See Fig. C.13.

Appendix D. Analysis of thermal and solutal Marangoni stresses

To physically justify the one-way coupling approach between the nitrogen concentration field and the melt pool fluid dynamics, a scale analysis of the free surface Marangoni stresses was conducted. The total Marangoni shear stress τ at the liquid-gas interface is the sum of its thermal and solutal components:

$$\tau = \frac{\partial \sigma}{\partial T}\nabla T + \frac{\partial \sigma}{\partial Y}\nabla Y \tag{D.1}$$

Assuming these gradients act over the same characteristic length scale along the melt pool surface from the keyhole to the trailing edge the relative magnitude of the stresses can be estimated by comparing the total absolute change in surface tension induced by each driving force ($\Delta\sigma_{thermal}$ and $\Delta\sigma_{solutal}$). For the solutal component, the surface tension σ

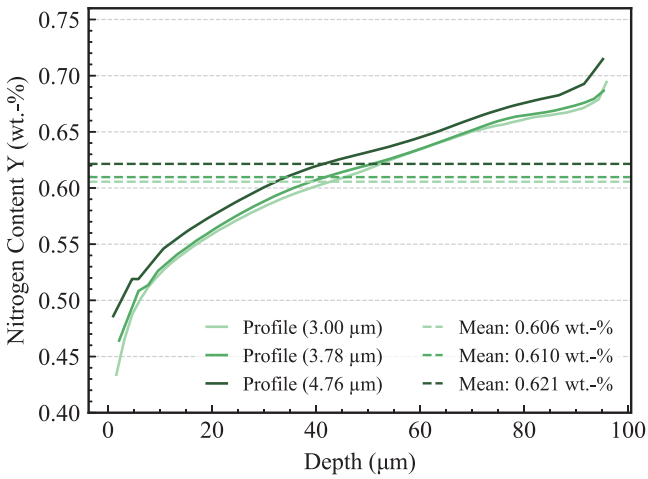


Fig. A.11. Grid convergence with the mean nitrogen concentration profiles for Y_0 = 0.75 wt-%, P = 150 W, v = 900 mm/s and Sh = 300.

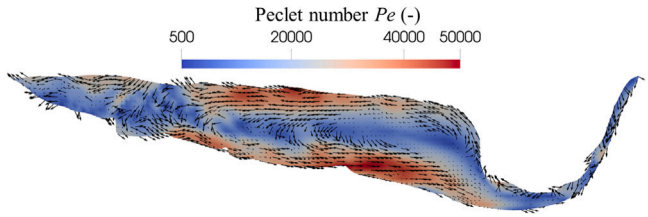


Fig. B.12. Pe number in longitudinal view of the melt pool for Y_0 = 0.75 wt-%, P = 150 W, v = 900 mm/s and Sh = 300 at t = 1.68 ms.

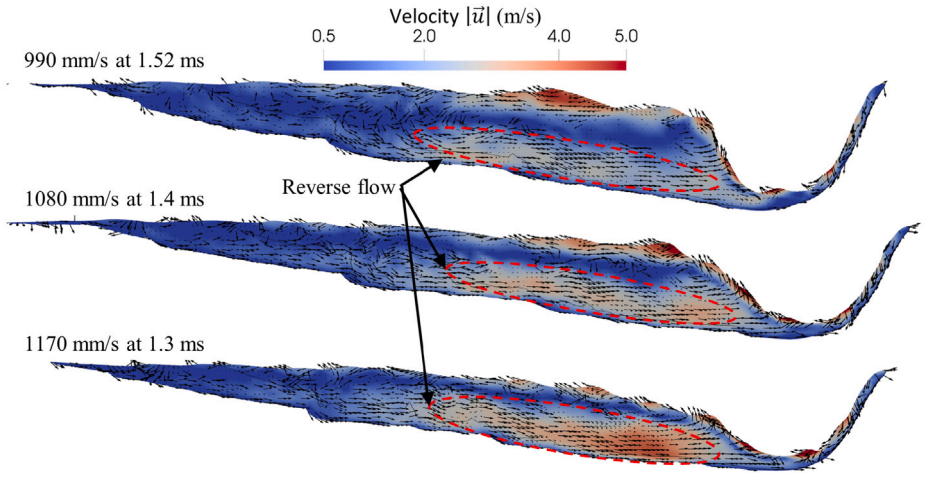


Fig. C.13. Longitudinal view of the melt pool for Y_0 = 0.71 wt-%, P = 150 W, Sh = 300 showing the magnitude of the velocity in combination with uniformly sized vectors for varying scanning speeds.

of liquid iron containing nitrogen can be evaluated using the following empirical relationship established by Zhu and Mukai [46]:

$$\sigma = 1930 - 267\ln(1 + 45a_N) \tag{D.2}$$

where a_N is the nitrogen activity. In the highly alloyed Fe-15Cr-0.3C-N system investigated in this study, the Wagner formalism yields a strongly suppressed activity coefficient of $f_N \approx 0.22$, primarily due to the high chromium content ($e^N_{Cr} = -0.047$). Within the maximum simulated concentration range of $\Delta Y = 0.25$ wt-% (from 0.75 wt-% down to 0.50 wt-%), the theoretical surface tension increases from approximately 1366 mN/m to 1458 mN/m. This yields a maximum solutal surface tension difference of $\Delta\sigma_{solutal} \approx 92$ mN/m. Assuming a linear relationship, this corresponds to a solutal coefficient of $\frac{\partial\sigma}{\partial Y} \approx -368$ mN/(m wt-%). For the thermal component, the temperature-dependent coefficient utilized in the model is $\frac{\partial\sigma}{\partial T} = -0.82$ mN/(m K). The free surface temperature drops from the peak keyhole evaporation temperature ($T_v \approx 3096$ K) down to the liquidus temperature (T_l = 1750 K) at the trailing edge of the melt pool. This creates a massive thermal gradient of $\Delta T \approx 1346$ K. The resulting thermal surface tension drop is calculated as:

$$\Delta\sigma_{thermal} = 0.82\ \text{mN/(m K)} \times 1346\ \text{K} \approx 1104\ \text{mN/m} \tag{D.3}$$

Comparing the two interfacial driving forces reveals a ratio of:

$$\frac{\Delta\sigma_{thermal}}{\Delta\sigma_{solutal}} = \frac{1104\ \text{mN/m}}{92\ \text{mN/m}} \approx 12 \tag{D.4}$$

This quantitative estimation proves that the temperature-driven Marangoni stress is more than an order of magnitude stronger than the concentration-driven Marangoni stress. Consequently, the macroscopic fluid dynamics and free surface deformations are predominantly governed by extreme thermal gradients, making the one-way coupling approach a physically reliable approximation for the investigated process regime.

Appendix E. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijheatmasstransfer.2026.129271.

Data availability

The developed code used in this study can be found at: https://zenodo.org/records/21410733. Further data will be made available on request.

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